Week 8 Assignment Solutions

Detailed step-by-step explanations for the uploaded 12-question PDF

Industrial Automation & Control

Week 8 detailed working

Built from the uploaded assignment PDF and lecture-topic request

Source used: the uploaded Week 8 assignment PDF with accepted answers visible in the pages. Each explanation below reconstructs the logic behind the correct option rather than only stating the final letter.

Question-solving approach used in this document

- **Step 1:** identify the concept being tested.
- **Step 2:** check each statement or option against the concept.
- **Step 3:** write one line for why the correct option works and one line for why the tempting wrong option fails.

Question 1. Reason for providing optocouplers inside PLC input/output units

Correct answer: Option (c): (i) False, (ii) True.

Step-by-step explanation

1. Statement (i) says optocouplers provide a fuse mechanism. That is not their main function. A fuse interrupts excessive current; an optocoupler transfers a signal using light and gives electrical isolation. So statement (i) is false.
2. Statement (ii) says optocouplers isolate the CPU from high voltages or currents. That is exactly why they are used in PLC I/O interfaces. Therefore statement (ii) is true.
3. Hence the correct combination is False/True.

Takeaway: Remember: fuse = overcurrent protection, optocoupler = isolation.

Question 2. Serial communication interface

Correct answer: Option (b): (i) True, (ii) False.

Step-by-step explanation

4. Serial communication transmits one bit at a time over the line, so statement (i) is true.
5. Parallel communication usually gives higher speed over very short distances because many bits move simultaneously. Serial is preferred for fewer wires and longer/noisy links, not because it is inherently faster in the general sense tested here. So statement (ii) is false.

Takeaway: Exam line: serial = one bit at a time; parallel = many bits at once.

Question 3. RS232 frame: purpose of extra bits X and Z at beginning and end

Correct answer: Option (c): (i) False, (ii) True.

Step-by-step explanation

6. In an RS232 asynchronous frame, the bits at the beginning and end are used for framing: start bit and stop bit.
7. These framing bits tell the receiver where the data word starts and stops, so statement (ii) is true.
8. They do not mainly check corruption during transmission. Error checking, if used, is done by parity or higher-level checks, so statement (i) is false.

Takeaway: Start/stop bits = framing, not parity.

Question 4. Bit Y in the shown RS232 frame

Correct answer: Option (c): (i) False, (ii) True.

Step-by-step explanation

9. The six data bits shown are 1100001. Count the number of 1s in the data: there are three 1s, which is odd.
10. If the system wants even parity, the parity bit must be 1 so that the total number of 1s becomes four, which is even.
11. Since the displayed bit Y is the parity location and the accepted answer marks only statement (ii) true, Y corresponds to even parity, not odd parity.

Takeaway: If data already has odd count of 1s, even parity requires parity bit = 1.

Question 5. Parallel data communication interface

Correct answer: Option (a): (i) True, (ii) False.

Step-by-step explanation

12. Parallel communication can send many bits at the same time, so over short distances it can be very fast. Hence statement (i) is true.
13. IEEE-488 is not the general statement they want for ordinary parallel interface comparison here; the accepted answer makes statement (ii) false in this context. Therefore the pair is True/False.

Takeaway: Parallel -> high speed, short distance, more wires.

Question 6. Communication over 100 to 300 m with high transmission rate

Correct answer: Option (c): (i) False, (ii) True.

Step-by-step explanation

14. RS232 is not suitable for 100-300 m high-rate industrial communication. It is mainly for shorter links and is more susceptible to noise over long runs. So statement (i) is false.
15. A 20 mA current loop is robust and is suitable for longer-distance industrial communication. So statement (ii) is true.
16. Hence the correct pair is False/True.

Takeaway: Long noisy distance: prefer current loop / differential industrial signaling, not RS232.

Question 7. Why flow control is the most important control in process industries

Correct answer: Option (b).

Step-by-step explanation

17. Flow directly affects how much mass or energy enters a process.
18. Because temperature, level, pressure and composition all depend on how fluids move, flow control indirectly influences those variables too.
19. Therefore option (b) is correct. The other options are too weak or plainly false.

Takeaway: Flow is the manipulated path behind many other loops.

Question 8. Valve that controls flow by rotating a ball with a hole through it

Correct answer: Option (c): Ball valve.

Step-by-step explanation

20. A ball valve contains a spherical ball with a bore through it.
21. When the ball rotates, the bore aligns or misaligns with the pipe and the flow changes.
22. That description exactly matches a ball valve, not globe, butterfly or diaphragm valve.

Takeaway: Quarter-turn + drilled ball = ball valve.

Question 9. Why installed flow characteristic differs from inherent characteristic

Correct answer: Option (d).

Step-by-step explanation

23. The inherent characteristic is defined when the pressure drop across the valve is kept constant.
24. In an actual piping system, as flow changes, pressure losses in piping and equipment also change. Therefore the valve pressure drop is not constant anymore.
25. Because of this varying pressure drop, the installed characteristic differs from the inherent characteristic. Hence option (d) is correct.

Takeaway: This is one of the highest-probability theory questions.

Question 10. Fastest actuator in valve operation

Correct answer: Option (a): Solenoid actuator.

Step-by-step explanation

26. A solenoid converts electrical energy directly to fast linear magnetic motion.
27. That makes it especially suitable for rapid on/off actuation.
28. Pneumatic and hydraulic actuators are powerful and common, but in the simple speed comparison asked here, the accepted answer is solenoid actuator.

Takeaway: Fast on/off -> solenoid.

Question 11. Why high-frequency command signals may not be followed accurately by valve stem movement

Correct answer: Option (b): Rate limits of the actuator.

Step-by-step explanation

29. A real actuator cannot move infinitely fast; it has finite velocity and acceleration capability.
30. When the command changes too rapidly, the stem cannot keep up, so tracking worsens.
31. That is a rate-limit problem. Noise or sizing may matter in other situations, but the direct cause asked here is actuator rate limit.

Takeaway: Finite actuator speed adds lag and tracking limit.

Question 12. Why equal percentage valves are preferred for valve sequencing applications

Correct answer: Option (c).

Step-by-step explanation

32. Equal percentage valves do not mainly eliminate hysteresis, guarantee tighter shutoff, or reduce actuator size.
33. Their main advantage is that equal increments of travel produce proportional percentage changes in flow capacity.
34. During sequencing, this helps avoid abrupt changes in overall loop gain when one valve hands over to another or when the operating point shifts widely.
35. Therefore option (c) is correct.

Takeaway: Equal percentage = smoother gain behavior over wide operating range.

Mini concept appendix used in the solutions

Concept	What to remember
Optocoupler	Isolation between field and logic side.
Serial	One bit at a time.
RS232	Short-distance serial link with start/stop framing.
Parity	Simple error-detection support, separate from start/stop bits.
Installed characteristic	Real system characteristic because valve pressure drop varies with flow.
Equal percentage valve	Good over wide operating range and sequencing.
Solenoid actuator	Very fast on/off action.
Actuator rate limit	Reason high-frequency commands may not be followed accurately.